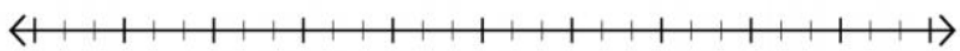
Grade 7 Name____________________
Unit 10 Date____________________
Lesson 2 Practice Period__________________

1. The asymmetrical property of inequalities states that if $a > b$ then $b < a$. Using this property, write an inequality that is equivalent to $x + \frac{1}{4} \leq 4$.

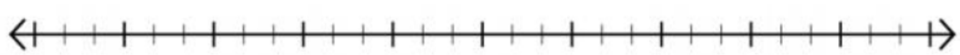
2. Complete the table to determine the value of $x - 6$ for each value of x.

$\boldsymbol{x}$	$20\frac{1}{2}$	19.25	18.5	$17\frac{1}{4}$	$16\frac{1}{2}$
$\boldsymbol{x-6}$					

3. For which values of x is the inequality $x - 6 < 12$ true? Represent the values from the table that are part of the solution set of $x - 6 < 12$ on the number line.



4. Graph the solution set for the inequality.



5. Brodhead Fitness charges a sign-up fee of $\$15$. Additionally, they charge a daily rate for visiting the gym. Francesca has no more than $\$32.50$ to spend on the gym for the month. Use the inequality $\$15 + g \leq \$32.50$ to represent the situation, where g stands for the daily cost of visiting the gym. Solve the inequality to determine the maximum amount of money that she can spend on daily visits for the gym for the month.

For questions 6-8, use the inequality $-28.5 < v - 16$ to complete the following:

6. Find the solution set by isolating the variable. Show your work.

7. Graph the solution set on the number line.

8. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

Solve each inequality. Then, graph its solution set.

9. $47\frac{3}{4} \geq w + 22$

10. $x + 8.2 < 18$